THE RELATIVE CYCLE PROPERTY SATISFIES H-DESCENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Functoriality of relative cycles under pullback (relative-cycles chapter, Lemma lemma-relative-cycle-functoriality, canonical tag 0H51). cited at proof lines 990-991. Base changes of relative cycles are relative cycles, proving the forward direction.
- The h covering notion and stability of h coverings under base change (More on Flatness, Definition flat-definition-h-covering, canonical tag 0ETS, and Lemma flat-lemma-h, canonical tag 0ETY). cited at statement lines 984-985 and proof lines 996-997. After replacing S by the valuation ring the pulled back family {S_i x_S S' to S'} is again an h covering.
- Refinement of qcqs V coverings (Topologies, Lemma topologies-lemma-refine-qcqs-V, canonical tag 0ETN). cited at proof lines 1002-1005. Since an h covering is a V covering, some member S_i carries points s' specializing to s lying over the generic and closed points of the valuation ring.
- Realizing specializations by valuation rings (Properties chapter, Lemma properties-lemma-locally-Noetherian-specialization-dvr, canonical tag 054F). cited at proof lines 1006-1008. Provides h : S' to S_i from a valuation ring mapping its generic and closed points to s' and s.
- Compatibility of specialization maps with base change along extensions of discrete valuation rings (relative-cycles chapter, Lemma lemma-specialization-extension, canonical tag 0H4K). cited at proof lines 1012-1015. The composite g = g_i composed with h is induced by a DVR extension, so the specialization identity transports from the base change to the original family.
- Injectivity of base change for cycles over fields (relative-cycles chapter, Section section-relative-fields, canonical tag 0H4E). cited at proof lines 1017-1018. The transported identity on the valuation ring base change descends because the closed-fibre base change map is injective.

Source reference. Original definition 0ETS: sources/stacks-chow/tags.txt, source lines 18527–18527. Cited at statement lines 984-985: the covering class in the statement.

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Source reference. Original lemma 0H51: sources/stacks-chow/tags.txt, source lines 21371–21371. Cited at proof lines 990-991: forward direction.

Source reference. Original lemma 0ETY: sources/stacks-chow/tags.txt, source lines 18533–18533. Cited at proof lines 996-997: the pulled back covering is again an h covering.

Source reference. Original lemma 0ETN: sources/stacks-chow/tags.txt, source lines 18523–18523. Cited at proof lines 1002-1005: extracts a member with the specialization over the two points of the valuation ring.

Source reference. Original lemma 054F: sources/stacks-chow/tags.txt, source lines 6638–6638. Cited at proof lines 1006-1008: realizes the specialization s' leading to s by a valuation ring map.

Source reference. Original lemma 0H4K: sources/stacks-chow/tags.txt, source lines 21355–21355. Cited at proof lines 1012-1015: transports the specialization identity along the composite.

Source reference. Original section 0H4E: sources/stacks-chow/tags.txt, source lines 21349–21349. Cited at proof lines 1017-1018: injectivity of the closed-fibre base change map lets the identity descend.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume S locally Noetherian and f locally of finite type. Let $r \geq 0$ be an integer. Let α be a family of r-cycles on fibres of X/S. Let $\{g_i : S_i \to S\}$ be a h covering (More on Flatness, Definition 0ETS). Then α is a relative r-cycle if and only if each base change $g_i^*\alpha$ is a relative r-cycle.*

Proof. If α is a relative r-cycle, then each base change $g_i^*\alpha$ is a relative r-cycle by Lemma 0H51. Assume each $g_i^*\alpha$ is a relative r-cycle. Let $g : S' \to S$ be a morphism where S' is the spectrum of a discrete valuation ring. After replacing S by S', X by $X' = X \times_S S'$, and α by $\alpha' = g^*\alpha$ and using that the base change of a h covering is a h covering (More on Flatness, Lemma 0ETY) we reduce to the problem studied in the next paragraph.

Assume S is the spectrum of a discrete valuation ring with closed point 0 and generic point η. We have to show that $sp_{X/S}(\alpha_\eta) = \alpha_0$. Since a h covering is a V covering (by definition), there is an i and a specialization $s' \rightsquigarrow s$ of points of S_i with $g_i(s') = \eta$ and $g_i(s) = 0$, see Topologies, Lemma 0ETN. By Properties, Lemma 054F we can find a morphism $h : S' \to S_i$ from the spectrum S' of a discrete valuation ring which maps the generic point η' to s' and maps the closed point $0'$ to s. Denote $\alpha' = h^*g_i^*\alpha$. By assumption we have $sp_{X'/S'}(\alpha'_{\eta'}) = \alpha'_{0'}$. Since $g = g_i \circ h : S' \to S$ is the morphism of schemes induced by an extension of discrete valuation rings we conclude that $sp_{X/S}$ and $sp_{X'/S'}$ are compatible with base change maps on the fibres, see Lemma 0H4K. We conclude that $sp_{X/S}(\alpha_\eta) = \alpha_0$ because the base change map $Z_r(X_0) \to Z_r(X'_{0'})$ is injective as discussed in Section 0H4E. □